



Australian Government

National Animal Health Diagnostics Business Plan 2021 to 2026: project updates November 2025

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Acknowledgement of Country

We acknowledge the continuous connection of First Nations Traditional Owners and Custodians to the lands, seas and waters of Australia. We recognise their care for and cultivation of Country. We pay respect to Elders past and present, and recognise their knowledge and contribution to the productivity, innovation and sustainability of Australia's agriculture, fisheries and forestry industries.

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Introduction

The [National Animal Health Diagnostics Business Plan 2021 to 2026](#) (NAHDBP) aims to further develop and strengthen the animal health diagnostics component of Australia's animal health system. Animal health professionals and diagnostic experts from governments, private laboratories and the university sector collaborated to develop the NAHDBP.

The NAHDBP guides the efficient and effective delivery of nationally coordinated activities to maintain and continually improve our national diagnostic capability and capacity. It is consistent with nationally agreed principles and objectives for biosecurity as defined under [Australia's Intergovernmental Agreement on Biosecurity](#) and, importantly, it complements the [Animalplan 2022 to 2027](#).

Responsibility for implementing activities under the NAHDBP resides with the Subcommittee on Animal Health Laboratory Standards (SCAHLs), with guidance and oversight from the Animal Health Committee (AHC) and funding identified through various sources.

This report provides an overview on the progress of each project linked to a NAHDBP objective. It summarises the project activities supported by the department. However, it does not cover all activities conducted under the NAHDBP from 2021 to 2026. Future reports may include activities across additional stakeholder groups.

For further information on the listed projects, email animalhealthlaboratories@aff.gov.au.

Objective 1

To enable high-throughput sequencing (HTS) usage and the application of HTS data to enhance animal disease investigations.

Table 1 Objective 1 projects

Project title	Lead and key collaborators	Description	Status	Progress update	Next steps
Australian Biosecurity Genomic Database (ABGD) for notifiable terrestrial animal viruses: Phase 1	- AgriBio (Victoria) - Department of Agriculture, Fisheries and Forestry (DAFF)	This project developed a verified genomic database of notifiable terrestrial animal disease viruses in Australia to facilitate a more robust and efficient application of HTS for Australian biosecurity surveillance and disease detection.	Complete	Nil	See: ABGD for notifiable animal pathogens: Phase 2
ABGD for notifiable animal pathogens: Phase 2	- AgriBio (Victoria) - DAFF	This project built upon ABGD Phase 1 by expanding the database to include verified sequences from viruses associated with Australia's national list of reportable diseases of aquatic animals. 10 new verified whole genome sequences from recent local terrestrial and aquatic outbreaks were uploaded improving utility of the database.	Complete	Nil	See: ABGD: Phase 3

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Project title	Lead and key collaborators	Description	Status	Progress update	Next steps
ABGD: Phase 3	- AgriBio (Victoria) - DAFF	This project built on previous phases of the ABGD project. Phase 3 aimed to add additional viral subtypes, expand to state based notifiable diseases, trial the inclusion of bacterial pathogens, and determine a long-term solution for database maintenance.	Complete	This project concluded in May 2025. The expanded ABGD Version 3 contains 202 reference genome sequences for viruses associated with diseases of terrestrial and aquatic animals in Australia that are notifiable or reportable at the national, state and territory levels. A manuscript was accepted into the journal <i>Database: The Journal of Biological Databases and Curation</i> .	A draft eLearning module is being further developed in collaboration with a software provider. Bioplatforms Australia has been engaged to host a standalone webpage for the ABGD project, which is currently under development.
Developing lumpy skin disease (LSD) and African horse sickness whole genome sequencing workflows	- CSIRO Australian Centre for Disease Preparedness (ACDP) - DAFF	This project developed robust whole genome sequencing procedures for both LSD virus and African horse sickness virus to mitigate the consequences of their outbreaks in Australia. The genomic sequencing probes and procedures, now available at ACDP, will be an invaluable resource in the event of an LSD outbreak in Australia.	Complete	Nil	Nil
Northern Australia biosecurity sequencing (NABSeq): high throughput sequencing (HTS) facility and network to enhance northern Australian biosecurity	- Berrimah Veterinary Laboratory (Northern Territory) - CSIRO ACDP - DAFF - Charles Darwin University	This project aims to improve northern Australia's ability to identify and respond to future biosecurity challenges; and provide more effective biosecurity risk management through modernisation of disease detection resources and surveillance in the region.	Extended	NABSeq continues to enhance northern Australia's HTS capability. The project has been extended to establish a sustainable business model, train project scientists on HTS workflows for emergency animal diseases (EAD) diagnosis and improve quality assurance for HTS capacity.	The next milestone report is due January 2026 and it is anticipated the project will be completed by June 2026.

Project title	Lead and key collaborators	Description	Status	Progress update	Next steps
Quality assurance for HTS as an infectious agent discovery tool	- CSIRO ACDP - DAFF	This project aimed to establish standardised workflows for the use of HTS in infectious agent discovery. It supported relevant national diagnostic HTS operating guidelines and improves overall quality assurance, allowing for high confidence when interpreting HTS results.	Complete	This project concluded in June 2025.	ACDP approved standard operating procedures (SOPs) will be submitted to SCAHLS for evaluation and national approval, and subsequent adoption by the Laboratories for Emergency Animal Disease Diagnosis and Response (LEADDR) network.
Development of a proficiency testing (PT) program for HTS bioinformatics within LEADDR – a roadmap	- CSIRO ACDP - DAFF	This project developed a roadmap for a PT program using diagnostic HTS and bioinformatics within the LEADDR network. This was achieved through conducting a planning workshop with LEADDR members to inform development of a business plan.	Complete	This project commenced in March 2025 and concluded in June 2025.	Nil

Objective 2

To scope and improve the national surge capacity of laboratory networks, including both government and non-government laboratories, during emergency animal disease (EAD) responses.

Table 2 Objective 2 projects

Project title	Lead and key collaborators	Description	Status	Progress update	Next steps
National laboratory simulation exercise planning	• SCAHLS • DAFF • All jurisdictional animal health laboratories	SCAHLS, through leadership of DAFF and an expert consultant, have completed the process of finalising a business plan to guide the roadmap for a national laboratory-focused simulation exercise. This exercise will assist in the identification of training and means by which to improve the surge capacity of Australian animal health laboratory networks in response to major EAD outbreaks.	Complete	Nil	See: National laboratory simulation exercise (Exercise Waterhole)

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Project title	Lead and key collaborators	Description	Status	Progress update	Next steps
National laboratory simulation exercise (Exercise Waterhole)	• SCAHLS • DAFF • All jurisdictional animal health laboratories	Following a scoping report completed in 2022, Exercise Waterhole included a series of discussion workshops over September and October 2023 and concluded with a functional exercise in early November 2023. The exercise activities tested and evaluated national laboratory preparedness for a major EAD incursion across the spectrum of laboratory services in Australia. A dual disease scenario was simulated involving a primary outbreak of LSD in northern Australia, and a concurrent outbreak of highly pathogenic avian influenza (HPAI) in southern Australia. Outcomes, including recommendations, also aimed to inform future planning for laboratory preparedness for and response to EADs.	Complete	The Exercise Waterhole report endorsed by AHC is now available at Exercise Waterhole - DAFF	DAFF will host a webinar on actions taken to address recommendations of Exercise Waterhole.
Building a sustainable national sample identification system for animal diagnostics - A pilot study using pre-barcoded sample collection tubes	• AgriBio (Victoria) • CSIRO ACDP • Gribbles VETLAB – Department of Primary Industries and Regions (South Australia) • DAFF	This project investigated pre-barcoded sample tubes to mitigate the bottleneck of manual labelling in disease outbreak responses. AgriBio conducted several pilot trials of selected sample tubes from the field to the laboratory. Trials were also conducted with the South Australian animal health laboratory service and ACDP to test its potential for use as a national system.	Complete	This project was delayed due to outbreak responses and was completed in May 2025.	SCAHLs will consider and assess a national laboratory pre-barcoded sample system based on the project output which will be developed by its recently formed working group.

Project title	Lead and key collaborators	Description	Status	Progress update	Next steps
Scoping a centralised national laboratory stockpile for emergency animal disease responses	- DAFF - SCAHLs - The LEADDR network	Exercise Waterhole highlighted the need for improved management of baseline or centralised stockpiles of laboratory supplied (Recommendation 16). In response, a National Laboratory Stockpile Workshop was held in March 2025 to explore a centralised stockpile for EAD responses. A scenario-based exercise was used to assess the requirements for a national stockpile.	Complete	The project commenced in February 2025 and concluded in March 2025.	SCAHLs and LEADDR continue to implement actions identified in the workshop as next steps.
Sample Tracking and Reporting System (STARS) enhancement	- CSIRO ACDP - DAFF	This project redeveloped the CSIRO-STARS network for enhanced biosecurity and laboratory management system interoperability at both national and jurisdictional levels. The expansion of software capability and improved performance to facilitate integration of a wider group of users to the network will improve Australia's ability to respond to EAD outbreaks.	Complete	Nil	See: National deployment of STARS v2: Phase 1
National deployment of STARS v2: Phase 1	- ACDP - AgriBio (Victoria) - Biosecurity Sciences Laboratory (Queensland) - Gribbles VETLAB - Department of Primary Industries and Regions (South Australia) - DAFF	This project seeks to deploy ACDP's STARS v2 in three jurisdictions. To achieve this ACDP will migrate current STARS nodes from the existing legacy STARS v1 client to STARS v2, assist jurisdictional staff to navigate integration and/or deployment of software, and provide implementation support by configuring software to each jurisdiction's application/security landscape.	On track	Implementation of STARS v2 has commenced in Queensland, Victoria and South Australia. The progress was also presented to SCAHLs for noting and discussion in September 2025.	The next milestone report is due November 2025, and the project is estimated to conclude in April 2026.

Objective 3

Introduce a suite of validated diagnostic assays to LEADDR laboratories targeting priority disease threats to Australian animal species.

Table 3 Objective 3 projects

Project title	Lead and key collaborators	Description	Status	Progress update	Next steps
Evaluation of antibody-detecting immunoassays for LSD in cattle and buffalo	- CSIRO ACDP - DAFF	This project evaluated the suitability of 3 antibody detection enzyme-linked immunosorbent assays (ELISAs) for LSD surveillance, and proof-of-freedom testing, in cattle and buffalo in Australia.	Complete	This project concluded in May 2025.	An ELISA kit developed with a commercial partner will be made available for use by Australian laboratories via the LEADDR network.
Establishing networked serological testing capability for African Swine Fever	- The LEADDR network - CSIRO ACDP - DAFF	This project established an improved serological testing capability for African swine fever in Australia by harmonising quality assurance processes and antibody-detection ELISA capability via the LEADDR network.	Complete	Nil	Nil

Project title	Lead and key collaborators	Description	Status	Progress update	Next steps
LSD testing capacity building in the LEADDR network	- The LEADDR network - CSIRO ACDP - DAFF	This project reinforced and extended Australia's LSD testing capability by enabling ACDP to participate in international PT. The project also facilitated robust and quality-assured national capability through implementation of PT and monitoring for LSD via the LEADDR Network.	Complete	The project was completed in June 2025, resulting in an established PT program for molecular — real-time quantitative polymerase chain reaction (qPCR) and serological — ELISA capabilities in the LEADDR network.	Nil
MicroRNA biomarkers for improved detection of animal diseases in a Johnes's disease (JD) model	- CSIRO ACDP - DAFF	This project evaluated the use of host microRNA for detection of JD in cattle and provided a proof of concept for the potential extension of this approach to other EADs.	Complete	This project was completed in January 2025.	See: Towards validation of a microRNA-based diagnostic test for JD
Towards validation of a microRNA-based diagnostic test for JD	- CSIRO ACDP - DAFF	This project further developed and validated molecular assays using host microRNA biomarkers to improve detection of JD.	Complete	This project was completed in May 2025. Progress included completion of WOA diagnostic validation stage 1-3 activities.	In consultation with stakeholders, CSIRO ACDP continue to seek support for a longitudinal study of JD in calves to further assess the molecular assay.
Developing immunohistochemistry (IHC) test for LSD	- CSIRO ACDP - DAFF	This Australian Centre for Disease Preparedness (ACDP) project developed and characterised an IHC test for the diagnosis and investigation of LSD virus.	Complete	Nil	Nil
Equine piroplasmosis (EP) diagnostic capability development	- CSIRO ACDP - DAFF	This project built ACDPs serological diagnostic capabilities for EP, caused by <i>Theileria equi</i> and <i>Babesia caballi</i> .	Complete	This project was completed in November 2024. Verification documents and a comprehensive serological testing algorithm have been developed.	See: Development of national equine piroplasmosis molecular diagnostic test capability

Project title	Lead and key collaborators	Description	Status	Progress update	Next steps
Development of national equine piroplasmosis molecular diagnostic test capability	- CSIRO ACDP - DAFF	This project will develop robust molecular procedures for detecting <i>T. equi</i> and <i>B. caballi</i> . These molecular diagnostic tests will complement the recently developed serological tests for equine piroplasmosis (EP) in horses at ACDP.	On track	A comparative analysis of available PCR assays has been conducted to identify and select the most suitable assay types for validation within this project's scope.	The next milestone report is due December 2025 and the project is anticipated to be complete by May 2026.
Improve national diagnostic capability for glanders	- CSIRO ACDP - DAFF	This project aims to build our national diagnostic capabilities for glanders, caused by the bacterium <i>Burkholderia mallei</i> .	Delayed	Development and optimisation of ELISA and PCR assay protocols is complete, along with acquisition of bacterial isolates for PCR assay validation.	The final milestone report outlining project delivery and evidence of accreditation documents for submission to National Association of Testing Authorities (NATA) is due October 2025.
Assessment and improvement of national bluetongue serological capability - a comparative study of commercial and in-house ELISAs	- CSIRO ACDP - DAFF	This project aimed to measure, assess, and improve on current bluetongue virus (BTV) serological capabilities in Australia to support national BTV surveillance and response activities.	Complete	The project was completed in June 2025. A manuscript outlining the complexities of BTV-group specific serology testing has been published the peer-reviewed journal <i>Viruses</i> .	Nil

Project title	Lead and key collaborators	Description	Status	Progress update	Next steps
Improvement of serological testing for flaviviruses in Australian livestock	• CSIRO ACDP • DAFF	This project aimed to develop a bead-based multiplexed serological (Luminex) assay for the differential diagnosis of various flaviviruses, suitable for use in pigs and horses. Successful uptake of this new serological approach at ACDP will enable faster, higher-throughput testing, and more sensitive/specific testing for flaviviruses in relevant livestock species.	Complete	This project concluded in October 2025. Significant progress was made to establish, optimise and assess the Luminex assay for the differential diagnosis of various flaviviruses in livestock, further development is ongoing at ACDP.	Nil
Assessment of molecular DIVA capabilities for LSD virus	• CSIRO ACDP • DAFF	This project aimed to assess existing molecular 'Differentiating Infected from Vaccinated Animals' (DIVA) assays that are based on real-time, or quantitative PCR, to determine if they are fit-for-purpose.	Complete	This project was completed in January 2025, with a DIVA testing algorithm recommended for roll-out to the LEADDR network.	Nil

Objective 4

Pilot the validation template process for point of care (POC) tests used for specific purposes.

Table 4 Objective 4 projects

Project title	Lead and key collaborators	Description	Status	Progress update	Next steps
Development and evaluation of a POC test network for EAD diagnosis	- EMAI (NSW) - DAFF	This project developed a suite of tests on a portable, battery-operated multiplexing qPCR platform for the detection of major EADs of cattle and pigs. It aimed to establish a proof-of-concept for multiplexing qPCR assays for emergency animal pathogens suitable for use under field conditions.	Complete	This project concluded in April 2025.	A finalised dossier outlining the validation approach used for POC tests shared through SCAHLS.
Consultancy for policies, strategies and operating guidelines for POC testing for infectious disease	DAFF	This project supports development of a nationally consistent management approach towards the use of POC tests specific for national notifiable and reportable diseases in terrestrial and aquatic animals in Australia.	Complete	Nil	Nil

Objective 5

Establish processes to develop and harmonise antimicrobial susceptibility-related diagnostic procedures across all relevant animal health laboratories

No DAFF supported projects address this objective.

Glossary

Term	Definition
ABGD	Australian Biosecurity Genomic Database
ACDP	Australian Centre for Disease Preparedness
AHC	Animal Health Committee
BTV	bluetongue virus
DAFF	Department of Agriculture, Fisheries and Forestry
DIVA	differentiating infected from vaccinated animals
EAD	emergency animal disease
ELISA	enzyme linked immunosorbent assay
EP	equine piroplasmosis
HPAI	high pathogenicity avian influenza
HTS	high-throughput sequencing
IFAT	immunofluorescence antibody test
IHC	immunohistochemistry
JD	Johne's disease
LEADDR	Laboratories for EAD Diagnosis and Response
LSD	lumpy skin disease
NABSeq	Northern Australia biosecurity sequencing
NAHDBP	National Animal Health Diagnostics Business Plan
NATA	National Association of Testing Authorities
PCR	polymerase chain reaction
POC	point of care
PT	proficiency testing
qPCR	quantitative polymerase chain reaction
SCAHS	Subcommittee on Animal Health Laboratory Standards
SOP	standard operating procedure
STARS	Sample Tracking and Reporting System